

Lecture 11: The First Law of Thermodynamics – Supplementary Notes

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These notes support Lecture 11. They contain background that was moved out of the contact time. Sections marked (extension) are not assessed.

1 The other sign convention

Some books write the first law as $\Delta U = Q + W_{\text{on}}$, where W_{on} is the work done *on* the system. The two conventions describe the same physics. Since $W_{\text{on}} = -W$, we have $\Delta U = Q - W$ in the convention of these lectures and of Young and Freedman. The mechanics sign (work done on a body) belongs to W_{on} ; the engineering sign (work done by the system) belongs to W . When you read another book, check which convention it uses before you copy a formula. In this course we use W for the work done *by* the system throughout, because the work of an engine (Lecture 12) is then positive.

2 Quasi-static processes and why the path exists

A gas has a single pressure p and temperature T only when it is in equilibrium. A process drawn as a line on a pV diagram is therefore a sequence of equilibrium states: it is *quasi-static*. In practice this means that the process is slow compared with the time the gas needs to equalise its pressure (about the time for sound to cross the container) and its temperature. A compression by a piston is quasi-static if the external pressure on the piston is, at every moment, only infinitesimally greater than the pressure of the gas; then $W = \int p \, dV$ uses the pressure of the gas itself. For a rapid compression, the gas next to the piston is at a higher pressure than the rest, and the work done on the gas is larger than $\int p \, dV$ evaluated with the average pressure. A quasi-static process with no friction can be run backwards through the same states and is called *reversible*; a free expansion cannot.

A process can be fast enough to be adiabatic and still slow enough to be quasi-static. This is the case for a sound wave in air (the compressions are too rapid for heat conduction to matter, but the pressure differences are small) and for the compression stroke of a good pump or engine.

3 Joule's experiment and the internal energy of an ideal gas

In Joule's free-expansion experiment a gas in one half of an insulated container expands into the evacuated half when the partition is removed. No heat enters ($Q = 0$) and, because nothing is pushed, no work is done ($W = 0$), so $\Delta U = 0$. For a dilute gas the temperature is found not to change. If U depended on the volume at constant T , then $\Delta U = 0$ with $\Delta V \neq 0$ would require a change in T . Therefore, for the ideal gas, U depends only on T .

(extension) For a real gas the effect is small but not zero. For the van der Waals gas, $U = nC_V T - an^2/V$, because the attraction between molecules adds a negative potential energy that becomes less negative as the gas expands. Setting $\Delta U = 0$ for a free expansion gives

$$\Delta T = \frac{an}{C_V} \left(\frac{1}{V_2} - \frac{1}{V_1} \right).$$

For 1 mol of nitrogen ($a = 0.137 \text{ Pa m}^6/\text{mol}^2$, $C_V = 20.8 \text{ J}/(\text{mol K})$) expanding from 10 L to 20 L, $\Delta T = (0.137)(50 - 100)/20.8 = -0.33 \text{ K}$: the gas cools slightly, because work is done against the attractive forces. The ideal-gas result $\Delta T = 0$ is therefore an excellent approximation at low density.

4 Work in the isothermal and adiabatic processes

For n moles of an ideal gas the isothermal curve is $p = nRT/V$ and

$$W_{\text{iso}} = \int_{V_1}^{V_2} \frac{nRT}{V} dV = nRT \ln \frac{V_2}{V_1}.$$

For the adiabatic curve $pV^\gamma = K$ (a constant),

$$W_{\text{ad}} = \int_{V_1}^{V_2} \frac{K}{V^\gamma} dV = \frac{K}{1-\gamma} (V_2^{1-\gamma} - V_1^{1-\gamma}) = \frac{p_1 V_1 - p_2 V_2}{\gamma - 1},$$

using $K = p_1 V_1^\gamma = p_2 V_2^\gamma$. This equals $nC_V(T_1 - T_2)$ because $C_V = R/(\gamma - 1)$ and $pV = nRT$. The result agrees with the first law, $W = -\Delta U$, which is a useful check.

5 A family of processes: $pV^k = \text{constant}$ (extension)

All four special processes are members of one family. For $pV^k = \text{constant}$ the same integral gives

$$W = \frac{p_1 V_1 - p_2 V_2}{k - 1} = \frac{nR(T_1 - T_2)}{k - 1},$$

and the heat is $Q = \Delta U + W = nC \Delta T$ with an effective molar heat capacity

$$C = C_V - \frac{R}{k - 1}.$$

The values of k are: $k = 0$ (isobaric, $C = C_V + R = C_p$); $k = \gamma$ (adiabatic, $C = C_V - R/(\gamma - 1) = 0$); $k \rightarrow \infty$ (isochoric, $C = C_V$); and $k = 1$ (isothermal, where $C \rightarrow \pm\infty$ because $\Delta T = 0$ while $Q \neq 0$). The family shows that C_p and C_V are not special properties of the gas, but the values of a general heat capacity along two particular paths. A process with $1 < k < \gamma$ has a negative C : the gas cools while it absorbs heat.

6 Heat capacities and the temperature (extension)

The values $C_V = \frac{3}{2}R$ and $\frac{5}{2}R$ hold over the range of temperature where the degrees of freedom are fully excited. Quantum mechanics shows that a motion is excited only when kT is comparable with the spacing of its energy levels. For a diatomic molecule there are three stages:

Temperature range (hydrogen, H_2)	Motions excited	C_V
below about 50 K	translation only	$\frac{3}{2}R$
about 200 to 1000 K	translation and rotation	$\frac{5}{2}R$
above about 3000 K	translation, rotation and vibration	approaches $\frac{7}{2}R$

The temperatures at which rotation and vibration switch on are characteristic of each molecule (about 85 K and 6000 K for hydrogen; about 3 K and 3400 K for nitrogen). This is why N_2 and O_2 behave as $f = 5$ gases at room temperature, and why vibration is ignored in the lecture.

7 Polyatomic molecules

A linear molecule such as CO_2 has three translations and two rotations; a non-linear molecule such as H_2O or CH_4 has three translations and three rotations ($f = 6$, $C_V = 3R = 24.9 \text{ J}/(\text{mol K})$, $\gamma = \frac{4}{3}$). The vibrational modes of polyatomic molecules have low energy spacings and are partly excited at room temperature. This is why the measured $C_V = 28.5 \text{ J}/(\text{mol K})$ for CO_2 in the table of the lecture exceeds $\frac{5}{2}R$, and why γ is smaller for more complex molecules.

8 Enthalpy (extension)

At constant pressure $Q = \Delta U + p\Delta V = \Delta(U + pV)$. The combination $H = U + pV$ is the **enthalpy**. It is a state function, and at constant pressure $Q_p = \Delta H$. For an ideal gas $H = U + nRT$, so $\Delta H = nC_p\Delta T$, which is the origin of the formula $Q_p = nC_p\Delta T$. Latent heats are changes of enthalpy at constant pressure: $L_v m = \Delta H$ in the boiling-water example of the lecture (6.77 kJ for 3.0 g), of which $\Delta U = 6.26 \text{ kJ}$ and $p\Delta V = 0.51 \text{ kJ}$.

9 The speed of sound (extension)

Sound in a gas is an adiabatic process with small changes in pressure. The speed is

$$v = \sqrt{\frac{\gamma p}{\rho}} = \sqrt{\frac{\gamma RT}{M}}.$$

For air at 20°C , with $\gamma = 1.40$ and $M = 28.8 \text{ g/mol}$, $v = \sqrt{(1.40)(8.314)(293.15)/0.0288} = 344 \text{ m/s}$. If the process were isothermal ($\gamma \rightarrow 1$) the result would be 291 m/s , which disagrees with the measured 343 m/s ; this was how Laplace showed that sound is adiabatic. Compare v with the rms speed of the molecules ($\sqrt{3RT/M} = 504 \text{ m/s}$, Lecture 10): the two are of the same order, as they should be since the disturbance is carried by the molecules.

10 A preview of Lecture 12

In the cycle of the lecture's integrated problem, 2000 J of heat was taken in and 331 J became net work; the remaining 1669 J was rejected. In tutorial problem B13 the ratio of net work to heat absorbed is $1 - 8^{-0.4} = 0.565$, depending only on the compression ratio. The first law allows any of these cycles, so long as $W_{\text{net}} = Q_{\text{net}}$. It does not tell us why the heat rejected can never be zero, or why heat never flows by itself from cold to hot. These restrictions are the content of the second law and of entropy, which are the subject of Lecture 12.